Reconstructing Meaningful Workflow from Transaction Logs

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***Abstract*—Enterprises today rely on workflow more and more to accomplish business goals. A well designed process can contribute to better business operation and working efficiency. In practice, however, most organizations have no efficient workflow, either there is not at all from the very beginning, or lack domain experts who equip with the comprehensive landscape about the business. In order to conduct business more effectively, it is necessary to reconstruct workflows from transaction logs, and further, optimize the business via meaningful representations of these workflows. In this paper, we propose a method and corresponding toolbox to mine the workflows and vital patterns. Domain experts can use them in an interactive way to reconstruct meaningful business processes. We got very promising experimental results on two real applications datasets**

***Keywords*—process mining; sequence mining; GSP; meaningful workflow**

I. Introduction

Enterprises today emphasize more on workflow that assemble tasks in order to accomplish business goals. A well designed workflow can contribute to better business operation and working efficiency. So, how to construct such workflows becomes a concern for business personnel. In practice, however, many organizations conduct their routine business without any precisely defined workflow. Even for those who have workflows designed in advance, business models will more or less deviate from their initials. In both situations, it is necessary for domain experts to have some effective means to reconstruct and optimize workflows.

New insights can be gained from past experiences. Fortunately, most enterprises save transaction logs for future auditing or achiving. These logs reflect faithfully the business operations traces, and provide us essential information for reconstructing and optimizing workflows. For instance,Table I and II respectively show a transaction log of an E-government application in China and a typical example workflow (ID = 1133917) logged in it. Business personnel are eager to find them out by means of some automatic tools. Furthermore, identifying vital patterns in large numbers of workflows can improve business significantly. Merging these two tasks can, to a great extend, shorten the execution period of workflows. The realize the dream, process mining is preferred.

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TABLE I
Transaction Logs in an E-Government Application

ID	Start Time	Entry
...	...	...
1132449	2009/6/15 11:09:24	Building trading acceptance
1133917	2009/6/15 11:09:24	Building mortgage acceptance
1133929	2009/6/15 11:09:25	Online registration application
1133929	2009/6/15 11:09:26	Registration approving
1133856	2009/6/15 11:09:33	Sealing examining
1132040	2009/6/15 11:09:39	Photocopying
...	...	...

TABLE II
Lifecycle of an Example Business Workflow

ID	Start Time	Entry
1133917	2009/6/15 11:09:24	Building mortgage acceptance
1133917	2009/6/15 11:10:51	Building mortgage examining and approving
1133917	2009/6/15 11:27:33	Sealing examining
1133917	2009/6/15 11:41:04	Certificate making
1133917	2009/6/15 14:17:11	Charging
1133917	2009/6/15 14:52:36	Certificate issuing
1133917	2009/6/15 14:55:38	Photocopying
1133917	2009/7/24 11:02:43	Filing

Process mining aims to discover information from transaction logs, and to rebuild and improve process models [1]. Mining tools like ProM [2] nowadays are also gaining more and more popularity due to its large quantity of mining algorithm as well as user-friendly interface.

However, some feedbacks from domain experts should not be overlooked in applying them to workflow reconstruction and optimization. First, the results obtained from many traditional process mining algorithms are elusory to business personnel. For example, one workflow from a real E-government application is illustrated in Fig. 4(a), domain experts reported that they cannot understand it very well, letting alone getting insight from it. Second, even some tools, e.g. ProM [2], can produce comprehensive workflow, they can hardly tell what patterns are vital to the workflow, which is expected to improve business; Third, some outliers in logs are mostly neglected by traditional mining tools due to the nature that they are overwhelmed by other frequent tasks. As we all know, less by no means implies meaningless. They should be identified and

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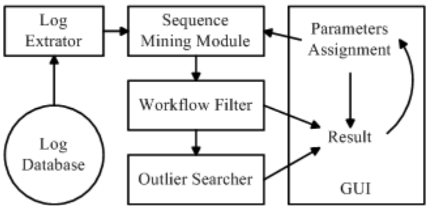

Figure 1. System Framework

presented to domain experts in order to reveal some knowledge about small probability events, either know-how or errors.

The work in this paper deals with above shortcoming of traditional process mining methods. It treats business processes as multiple execution threads. So here, we can take advantage of GSP [3], a sequential pattern mining algorithm to advise domain experts: (1) what kind of processing patterns can be extracted as workflows; (2) which sequential patterns are most frequently occurred in workflows; (3) what kinds of outliers can be affiliated to resulting workflows. By discovering these three factors, business people can not only reconstruct their workflow from logs, but also optimize existing workflows more easily.

Our work embodies its technical contributions in two folds:

1) A process mining method which can present meaningful results to domain experts;
2) Exposing vital patterns (both frequent ones and accidentals) affiliated to workflows.

These benefits are demonstrated by applying our method to two real business transaction logs.

II. Framework

The framework of our system is presented in Fig. 1. The log database maintains routine task instances with time-stamps (Table I has shown a partial snapshot of the log). By reading transaction log database, log extractor will rearrange log files into threads just like the ordered tasks sequences in Table II. Then sequence mining module discovers workflow candidates according to the parameters assigned by domain experts. Successively, workflow filter and outlier searcher will polish each result pattern with affiliated tasks. Finished workflows are presented to users. With these results, domain experts can interactively adjust parameters, in order to obtain workflows that meet business criteria.

III. Design and Implementation

Before technical details, it is necessary to define some relevant concepts.

A Sequence $s = \langle \tau_1, \tau_2, ..., \tau_l \rangle$ is an ordered list of tasks with length of l. A task τ_i is an activity in a workflow. Usually, a task τ_i should have a timestamp (denoted as $\tau_i.t$) to record when it was start to execute. Generally, $\tau_i.t \leqslant \tau_j.t$ if $1 \leqslant i \leqslant j \leqslant l$.

In our example, the activities in Table II can be regarded as a sequence: $s_{1133917}$ = ⟨Building mortgage acceptance, Building mortgage examining and approving, Sealing examining, Certificate making, Charging, Certificate issuing, Photocopying, Filing⟩.

Sequence $s' = \langle \tau'_1, \tau'_2, ..., \tau'_k \rangle$ is called a *sub-sequence* of $s = \langle \tau_1, \tau_2, ..., \tau_l \rangle$, or s is said to *contain* s', if $\{\tau'_1, \tau'_2, ..., \tau'_k\} \subseteq \{\tau_1, \tau_2, ..., \tau_l\}$ with the same order.

For example, s' = ⟨Charging, Photocopying, Filing⟩ is a sub-sequence of $s_{1133917}$. And $s_{1133917}$ contains s'.

A. Support and Confidence

In data mining, *support* and *confidence* are two essential criteria to reveal the relationship of data. In our study, we define them as follows.

Support: if **S** is a set of sequences, i.e. **S**=$\{s\}$, for all sequence $s' \in$ **S**, The *support* of s in **S**, say s_{supp}= P(s is a sub-sequence of s'). In another word, *support* denotes the probability that a sequence occurs in workflows. *Support* gives us the quantitative definition of the term "frequency".

For example, consider a sequence ⟨Photocopying, Filing⟩ which appears in the workflow of Table II. If it occurs totally 10 times among all 100 workflows in Table I, the *support* of this sequence is $10/100 = 0.1$. Usually, support is compared with a threshold given by users, called *minimum support*. If it is no less than the *minimum support*, we call this sequence "sequential pattern".

Confidence: if **S** is a set of sequences, i.e. **S**=$\{s\}$, for all sequence $s' \in$ **S**, the *confidence* of s in **S** , say s_{conf} = P ($s = s' | s$ is a sub-sequence of s'), $s' \in$ **S**. So, *confidence* is the conditional probability that a sequence s is the same as another sequence s', under the condition that s is a sub-sequence of s'.

For example, given **S** and a sequence s, if in **S**, only s_1, s_2 and s_3 contain s and $s_1 = s, s_2 = s, s_3 \neq s$. We can say that the *confidence* of s in **S** is 0.67 (= 2/3).

B. GSP - A sequential pattern mining algorithm

GSP is a sequential pattern mining algorithm developed by R Srikant *et al.* in 1996 [3]. The algorithm continues to construct, trim and find sequential patterns of length i by using sequential patterns of length $i-1$ until no sequence's support is bigger than the minimum. Its objective is to find dominant sequences. We use the GSP as a base mining engine in our study, but the output of GSP will then be translated into meaningful workflows, as described below.

C. Mining and Filtering Algorithm

After mining out all candidate sequential patterns using GSP, it is necessary to polish these sequences into useful ones.

The first thing to do is to eliminate redundant sequences from candidate sequential patterns. Given a sequence set **S**, and two sequences s and s' which satisfy $s, s' \in S$. If s' is a sub-sequence of s, we may regard s' as a redundant sequence which should be removed from **S**. This step is important because if a sequence of length 8 is a candidate sequential pattern, all of its 35 ($= \sum_{i=2}^{8} i$) sub-sequences will also be candidates. If we return results without trimming, end users may be overwhelmed by less informative data. The filtering algorithm is outlined in Alg. 1.

However, a trimmed candidate sequence set may be still too large for Users to grasp. So, another parameter *RoC* (range of

Algorithm 1 Workflow Filtering

Require: sequential pattern set seqPatSet
Ensure: sequential pattern set Result
  sort seqPatSet order by sequence length desc
  move the first sequence in seqPatSet to Result
  for all sequence seqPat in seqPatSet **do**
    for all sequence s in seqPatSet **do**
      if seqPat is not a sub sequence of s **then**
        add seqPat to Result
      end if
    end for
  end for

Algorithm 2 Outlier Searching

Require: sequence set seqSet ∨ sequential pattern seq ∨ sequence set outlierSet
Ensure: sequential pattern set Result
  for all sequence seqPatCand in seqSet **do**
    if seq is a sub sequence of seqPatCand∧ seqPatCand≠ seq **then**
      add seq to outlierSet
    end if
  end for
  Result ⇐ GSP_mining (outlierSet)

confidence) is needed. The *RoC* classifies sequential patterns into different groups. Only those sequences whose confidence is within the given interval should be returned. For example, if an end user is willing to find out what are the most frequent workflows in the logs, he may set the range of confidence close to 1,say [0.9, 1]. Thus the returned sequential patterns, to a great extend, are real workflows. However, if an end user wants to discover the common knots in most workflows, he may set the range of confidence close to 0, say [0, 0.1]. The system will provide him results with sub-sequences of workflows.

D. Searching for Outliers

Sometimes, business personnel will also be keen on finding some small probability events, called outliers. In most frequency-based mining algorithms, outliers are regarded as noise that should be eliminated. But from the feedback from domain experts, they want to probe into these outliers to gain know-how, or to find the flaw in business practices. What we need is a smart treatment of such outliers.

In this context, when a sequence pattern we find is close to a real workflow, its outlier tasks are assumed to be attached to the sequence to handle some unexpected situations such as rollback or further charging. With this concern, we know that the length of an outlier sequence is longer than the normal one. The main idea to find outliers of a sequence is again relied on the GSP algorithm. The mining target is a set of sequences **S** which satisfies: for all $s' \in \mathbf{S}$, where target sequential pattern s is a sub-sequence of s' and $s \neq s'$. Then the mining algorithm will return outliers affiliated to corresponding workflows. The outlier searching algorithm is shown in Alg. 2.

IV. Experiments and Discussion

To justify our method, we built a prototype(can be got on request.) using Java and GraphViz [4].

A. Dataset

We employed two real datasets to test the system. One is a one year log from an E-government application in China. It contains 656952 entries, 150853 instances of processes and 568 different tasks. Since the application is process-aware, we can verify reconstructed workflows against original process definitions. Another dataset is a one year routine activities collected from a hospital. It contains 52237 entries, 11702 instances of processes and 18 different tasks. No any process defined in advanced. Doctors want to get some insight about business process from the logs.

B. Experiment Results

Fig. 2 shows mining results on the E-government's dataset. In order to give more informative result, we use arcs with different width proportional to the support indicators. The thicker the arc presents, the larger the support is. The sequential pattern of the graph on the left means ⟨Building mortgage acceptance, Building mortgage examining and approving, Sealing examining, Certificate making, Charging, Certificate issuing...⟩ should be identified as a workflow, so does the right one.

Due to the layout concern, only 2 out of 5 sequential patterns are shown in Fig. 2. The solid arcs in the graph represent regular sequential patterns and dotted ones are outlier tasks. All of the regular sequential patterns occur at least 3% among all workflows and 95% of which are real workflows. So business personnel may regard these workflows as frequent ones and optimize their structures. Meanwhile, the abnormal tasks here may be regarded as rollback activities. Checking against given process definitions, domain experts are highly satisfied with the functionality.

Fig. 3 shows a mining result on the hospitals dataset, the support is 0.03, confidence ranges from 0.1 to 0.2. Numbers in boxes are tasks IDs. It seems trivial, but reflects the simplicity nature of ambulatory treatment in hospitals. To serve as a tool to reconstruct and optimize work.

C. Compare with ProM

Fig. 4 shows a comparison between our system and ProM in terms of meaningfulness. The experiment dataset is the same as the E-government one. Fig. 4(a) is the process generated by using Alpha [5] algorithm in ProM [2]. Apparently, it is too heavy and complicated to understand, even for its part depicted as Fig. 4(b) . Since the plug-in fails to support the Chinese characters, we use task IDs to denote different tasks. Fig. 4(c) shows the result of our system.

From the feedbacks from domain experts, it is almost impossible for to understand what Alpha Algorithm mines out. The mass relationship among tasks made the whole graph

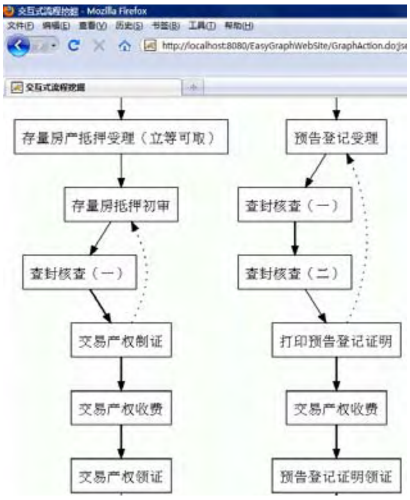

Figure 2. Mining result on e-government's dataset. Support = 0.03, confidence = [0.95, 1], outlier sequence enabled.

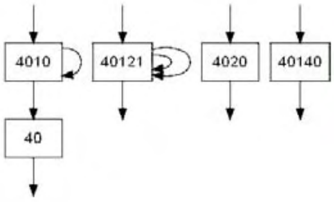

Figure 3. Mining result on hospital's dataset. Support = 0.03, confidence = [0.1, 0.2] , outlier sequence disabled.

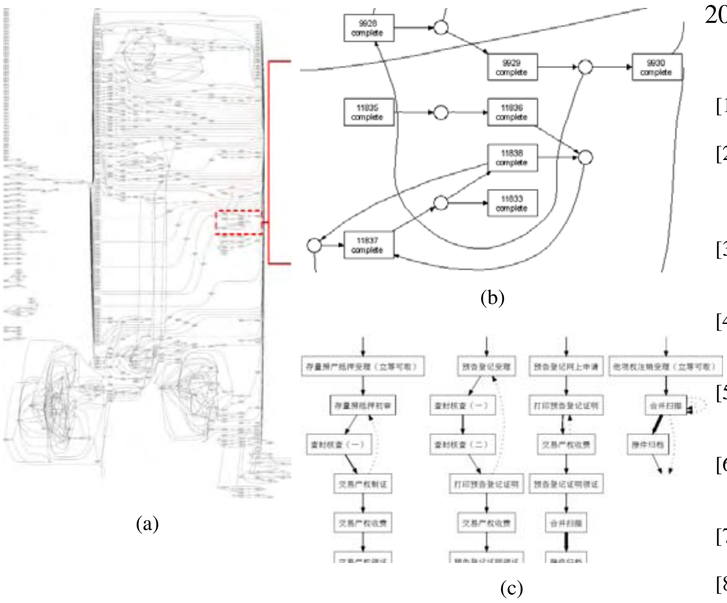

Figure 4. Comparison between the graph of our result and the one generated by Alpha mining algorithm.

meaningless. Contrast to it, in our system, each mined process can be mapped to one or a group of meaningful workflows. With such meaningful process, domain experts can improve their workflows easily.

V. Related Work

Process mining has been extensively researched [6], [7], [8] over the last two decades, which was surveyed by van der Aalst et al in [1]. In concurrency, the alpha algorithm [5] made the first step. Under the framework of ProM [2], more techniques [5], [8] were approached.

Sequence mining [9] is a part of process mining. Many works of it are also under research [7]. The study of sequential pattern mining basically started from the traditional data mining. R Agrawal et al. proposed AprioriAll, AprioriSome and GSP [3] methods on the base of a classic data mining algorithm Apriori. Then M. Zaki et al. also proposed SPADE [10] as a Apriori-based method. J Pei, et al continue to work on this field and PrefixSpan [11] ran as a representative.

VI. Conclusion

In this study we explored an interactive way for business personnel to discover meaningful workflows from transaction logs. We suggest using sequential pattern mining method to find the frequently used workflows or most common subsequences which can be fundamental for process optimization. In order to look for abnormal or affiliated tasks in workflows, we also developed a brand new method to polish workflow candidates. Experiments against two real datasets are very promising, especially for the meaningfulness factors.

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